





Design, Fabrication, and Integration of a Soft Pneumatic Gripper

Internship Report

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November 1, 2026

Acknowledgement

We would like to express our sincere gratitude to the Medical Ultrasound Research Laboratory (MURL) and the Soft Matter Group (SMG), Indian Institute of Technology Hyderabad for providing us with the opportunity to undertake this internship and gain valuable research experience in the field of soft robotics. This internship has significantly enhanced our technical knowledge and practical skills. We are deeply grateful to our supervisors Dr. Alan R. Jacob and Dr. Avinash Eranki for their constant guidance, encouragement, and valuable suggestions throughout the project. Their expertise, support, and insightful discussions helped us successfully carry out the research and the guidance helped us overcome the challenges we faced. We are also thankful to Dr. Suhanya Duraiswamy for guidance and access to Digital Light Processing printer which enabled us to manufacture the adapter that mounts the end-effector on the robotic arm. We also extend our heartfelt thanks to all the research scholars and laboratory members of the MURL and SMG Labs for their technical assistance and support during material preparation, fabrication, experimentation, and testing. Their guidance made our learning experience more meaningful and productive. We sincerely acknowledge the excellent laboratory facilities and research environment provided by IIT Hyderabad, which enabled us to perform the experimental work efficiently. Finally, we would like to express our heartfelt gratitude to our family and friends for their constant encouragement, motivation, and unwavering support throughout this internship. Their confidence in us inspired us to complete this project successfully.

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1 Introduction

Soft robotics is an emerging field of robotics that focuses on the development of robots using soft, flexible, and compliant materials instead of conventional rigid components. Inspired by biological structures such as the human hand, octopus tentacles, and elephant trunks, soft robotic systems possess the ability to safely interact with delicate objects. Their flexibility, lightweight nature, and compliance allow them to inherently yield and conform to the physical contour of the target object which makes them highly suitable for applications where conventional rigid robots are limited.

Among the various soft robotic technologies, soft pneumatic grippers have attracted significant attention due to their simple design, adaptability, and ability to grasp objects of different shapes, sizes, and surface textures without causing damage. These grippers operate by supplying compressed air into internal pneumatic chambers, producing controlled bending and motion of the soft actuator. This mechanism enables stable and gentle grasping, making soft pneumatic grippers suitable for applications in biomedical engineering, healthcare, food handling, agriculture, industrial automation, and human–robot interaction.

The performance of soft pneumatic grippers depends on several factors, including material properties, actuator geometry, fabrication techniques, and pressure control. Silicone-based elastomers are commonly used because of their excellent flexibility, elasticity, durability, and biocompatibility. The fabrication process generally involves material preparation, mould design, degassing, casting, curing, and performance evaluation to ensure reliable and repeatable operation.

With continuous advancements in materials science, manufacturing techniques, and robotic control systems, soft pneumatic grippers have become an important area of research for developing safe, adaptive, and intelligent robotic manipulation systems. Their ability to handle fragile and irregularly shaped objects makes them a promising solution for a wide range of future robotic applications.

Literature Study

The design and fabrication of soft pneumatic actuators (SPAs) heavily rely on the principle of asymmetric expansion, where pressurized air causes internal chambers to inflate unevenly against a strain-limiting bottom layer, resulting in controlled bending [1]. Material selection is critical to optimizing this actuation. While commercial silicones are common, utilizing a custom blend of Siloczest Liquid Silicone Rubber (LSR) mixed with 10 wt% of 10 cSt PDMS silicone oil provides a highly favorable balance of viscosity and mechanical strength [6]. This custom formulation reduces overall material costs by approximately 30% compared to standard commercial alternatives, however the drawback is that excessive oil addition can reduce the structural strength and slow down curing times [6]. Furthermore, the internal geometry of the actuator significantly influences its characteristics - rectangular chamber profiles featuring thinner walls maximize both the deflection angle and the output force while requiring less actuation pressure [4].

Experimental characterization of these soft grippers demonstrates that both the bending angle and the mass-lifting capacity increase proportionally with the applied internal pressure [7, 8]. Recent advancements in soft robotics have introduced multi-degree-of-freedom (multi-DOF) actuators capable of highly dexterous spatial manipulation, that work by selectively pressurizing specific combinations of internal chambers. These mod-

ules can generate complex, coupled motions that result in multidirectional bending [9,10].

2 Material Selection

The material properties directly influence its flexibility, durability, and actuation performance. During this project, different silicone elastomers were evaluated during the fabrication process. The recommended material used by the original research for manufacturing the PneuNet SPA is Ecoflex 0030 [3]. However, a study [6] found a cheaper alternative custom blend of Siloczest LSR and Polydimethylsiloxane (PDMS) silicone oil that performs similarly to Ecoflex. The general trend observed was that PDMS oil acts as a diluent that lowers the mixture's viscosity, aiding in filling intricate molds. However, increased oil addition leads to reduced overall strength and slower curing times. Among the tested formulations, the LSR blend containing 10 wt% of 10cSt silicone oil provided the most favorable balance of mechanical properties and viscosity. Additionally, this custom blend proved to be roughly 30% cheaper than other commercially available soft materials. In our manufacture of soft gripper we decided to use solely the Siloczest LSR (shore hardness 10 A), without the addition of external diluents as a first iteration and Ecoflex was used for the manufacture of the final grippers.

2.1 Materials and Components Used

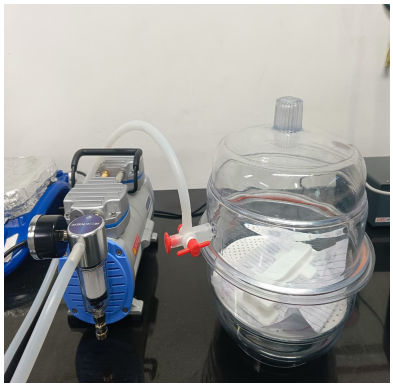
- **Siloczest Liquid Silicone Rubber (LSR):** Used for initial fabrication trials, process optimization, and prototyping because of its good curing characteristics, ease of processing, and economical cost.
- **Ecoflex:** Used for fabricating the final soft pneumatic gripper fingers due to its excellent flexibility, high elasticity, and superior bending performance under pneumatic actuation.
- **Silicone Oil:** Added to the silicone mixture to reduce viscosity, improve material flow during casting, and increase the flexibility of the cured actuator (however we did NOT use this during our prototyping).
- **Cross-linker (Curing Agent):** Mixed with the silicone elastomer in the required proportion to initiate curing.
- **3D Printed Mould:** Used to fabricate the soft gripper mould design.
- **Vacuum Desiccator:** Used to remove entrapped air bubbles from the silicone medium, and also to boil off low molecular weight impurities.
- **Centrifuge:** Assisted in removing residual bubbles and improving the homogeneity of the silicone-crosslink mixture.
- **Syringe and Pneumatic Tubing:** Provided compressed air for actuating and testing the soft pneumatic gripper.
- **RLDP-5 Robotic Arm:** Used to integrate and evaluate the fabricated soft gripper for robotic grasping applications.
- **Unitree L2 LiDAR:** Used for object detection in the environment.
- **Miscellaneous Laboratory Equipment:** Mixing containers, weighing balance, syringes, spatulas, and measuring tools were used for accurate material preparation and fabrication.

2.2 Fabrication Process

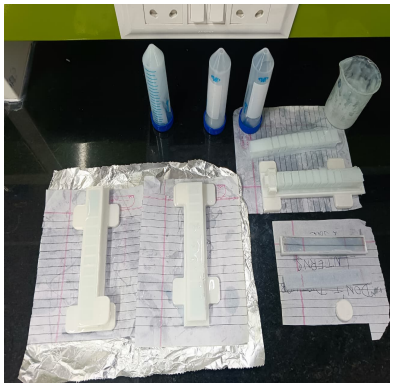
2.2.1 Initial Process and Challenges Faced

In our initial prototyping, we followed the resource provided by the Soft Robotics Toolkit for manufacturing the PneuNet actuator [2]. Our initial procedure was as follows:

- **Prepare the Elastomer:** Mix the Siloczest LSR components thoroughly in a 100:3 ratio
- **Pour Elastomer into Moulds:** Fill the main body mould completely with the mixed elastomer and the base mould only halfway
- **Add Strain-Limiting Layer:** Place a piece of paper on top of the liquid elastomer in the base mould to serve as an inextensible strain-limiting layer.
- **Initial Curing:** Cure both moulded parts separately. (This process can be accelerated using an oven)
- **De-mould Main Body:** Once fully cured, carefully remove the main body piece from its mould
- **Apply Bonding Layer:** Fill the remaining half of the base mould with freshly mixed, uncured elastomer.
- **Assemble the Actuator:** Place the cured main body onto the base mould. The uncured elastomer will act as a glue to bond the main body piece to the base.
- **Final Curing:** Cure the two assembled pieces together.
- **De-mould Final Actuator:** Remove the completely sealed actuator from the base mould.
- **Connect Air Source:** Attach the required tubing and air source to pressurize and test the soft actuator.



(a) De-gassing step inside desiccator



(b) Final curing process (could be heated at 60° C)

Figure 1: Steps during manufacture of SPA

However this process was not ideal due to the formation of a large number of bubbles throughout the liquid medium after pouring into the mould. Our initial approach to this was to degas the mould after pouring the liquid silicone, and take it out and pop the emergent bubbles on the surface of the mould for 2-3 cycles of 10 minutes. A majority of the bubbles could be removed via this process, but it did not guarantee complete removal of bubbles. The bubbles that were trapped below the ridge separating the chambers within the mould were also inaccessible and therefore would form weak points at the thinnest regions of the finger. Further, when degassing, the trapped bubbles would increase in volume while getting expelled from the liquid mould, resulting in leakage of the liquid material from the top. Yet another potential cause for the bubbles was that when degassed under low pressure, some low molecular weight impurities within the LSR would boil off, causing non-uniformity in the fluid. All of these would result in these trapped bubbles acting as weak-points post-curing and bursting when applying air pressure. This would render the actuator unusable.



(a) (b)

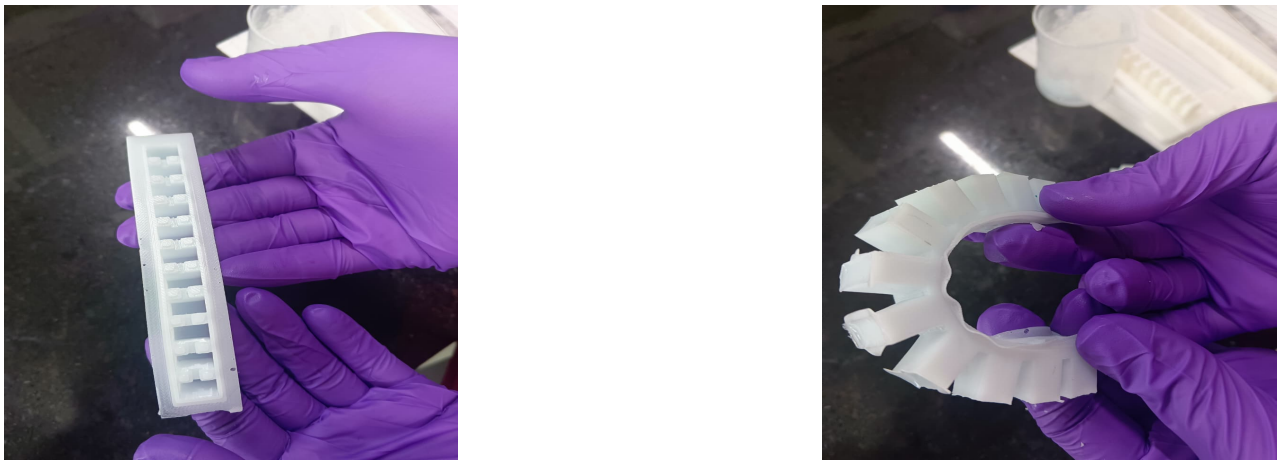
Figure 2: Defects in SPA due to bubble formation in the ridges between the chambers

2.2.2 Improved Process

Therefore, to overcome the above challenges, a controlled manufacturing pipeline is proposed to address excessive bubble formation and improve uniformity:

- **Pre-Desiccation:** Part A of the Liquid Silicone Rubber (LSR) is placed in a vacuum desiccator for 4–5 hours before adding cross-linker to remove low-molecular-weight impurities and reduce bubble formation during the fabrication process.
- **Controlled Mixing:** To minimize the introduction of atmospheric air, all mixing procedures were then conducted inside a fume hood. The corresponding cross-linker (Part B) is added, and the mixture is blended using an overhead stirrer rather than by hand, ensuring a highly uniform consistency.
- **Centrifugation:** The blended mixture is then immediately centrifuged at 7500 RPM for 5 minutes. This forces any residual trapped bubbles to the surface and ensures complete homogeneity. Since the working pot life after adding the cross-linker was approximately 45 minutes, the subsequent steps are performed rapidly.
- **Casting and Final Degassing:** The prepared silicone mixture is carefully poured into the 3D-printed mould and placed in a vacuum desiccator for 10–15 minutes to remove any remaining trapped air bubbles.
- **Curing:** The mould is left undisturbed for 4–5 hours to allow complete curing of the silicone, after which the fabricated soft pneumatic gripper is carefully demoulded

for further testing and integration.



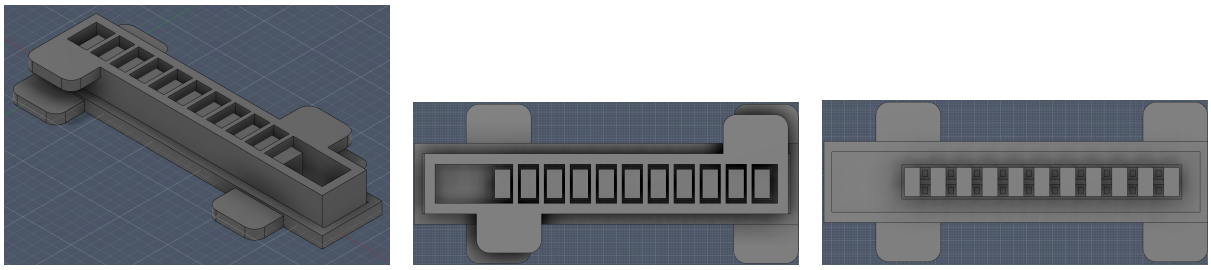
(a) Interior chambers of SPA (b) Final product

Figure 3: Final soft pneumatic actuator

3 Mould Design and Fabrication

3.1 Mould Design

The mould design that we used was a modified version of the PneuNet actuator from the Soft Robotics Toolkit [5]. Polylactic acid (PLA) plastic was used to manufacture the mould via 3D printing. It consisted of an upper and a lower part in order to facilitate easy pouring of silicone, proper alignment, and demoulding after curing. Care was taken to ensure that the internal air channels were formed precisely, enabling uniform inflation and controlled bending of the gripper.

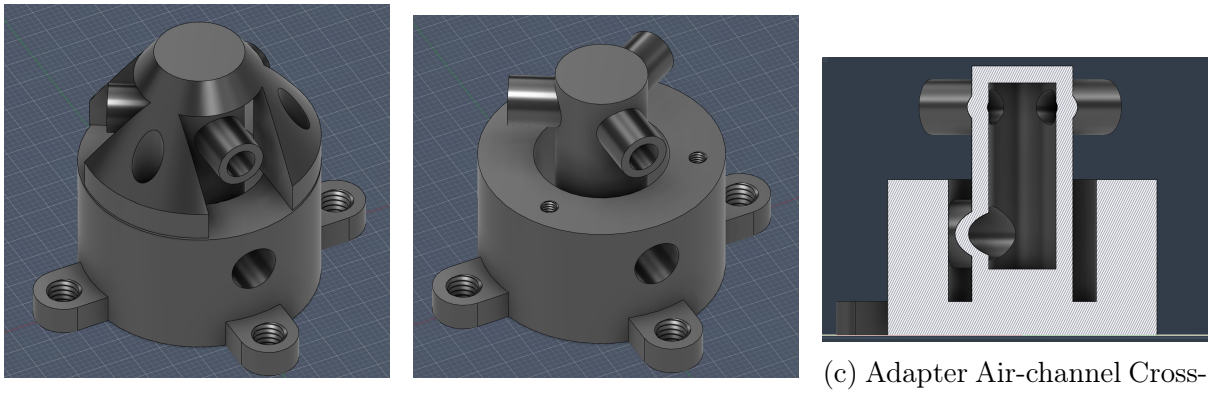


(a) (b) (c)

Figure 4: SPA Mould Design

3.2 Mould Assembly

Before casting, the upper and lower mould components were cleaned and assembled securely. To prevent potential leakage of silicon during pouring, a small layer of the LSR was used to seal the interface between the two parts with a spatula. After curing, the mould was carefully disassembled, and the fabricated gripper was removed without damaging the silicone structure.

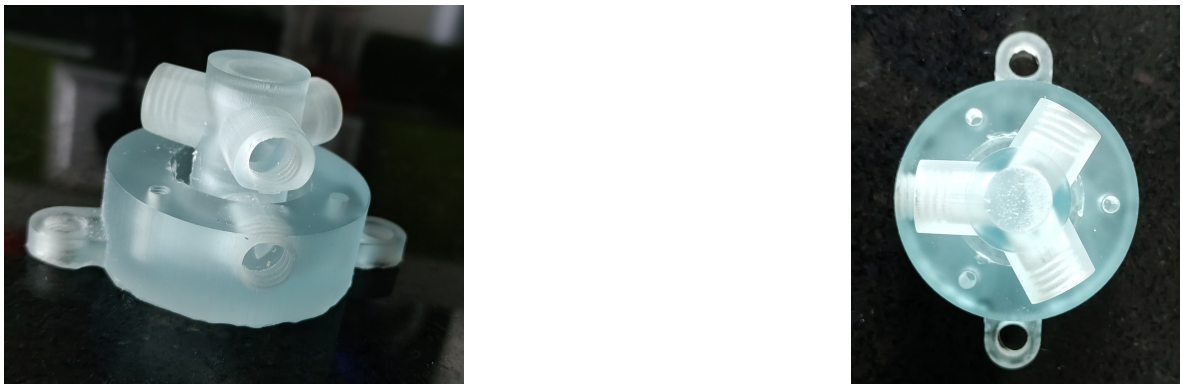


(a) Modular Finger Adapter (b) Finger Adapter Lower Base (c) Adapter Air-channel Cross-Section

Figure 5: End-effector to arm adapter design

4 Actuation Mechanism and Hardware Iterations

The performance of a soft pneumatic gripper depends not only on the actuator design but also on the pneumatic control system and hardware integration. During the development process, several iterations were carried out to improve the actuation mechanism, pressure control, and air inlet design, resulting in a more reliable and efficient gripper.



(a) Side View (b) Top View

Figure 6: Resin-printed Adapter

4.1 Precise Pressure Control

The fabricated soft pneumatic actuator was actuated by supplying compressed air through a syringe during the initial testing phase. It was observed that when a controlled volume of air was introduced into the actuator, the finger bent gradually and reached a stable equilibrium position. Thus, precise bi-directional control is required to fully actuate (curl) and retract (extend) the finger. Positive air pressure is required to bend the fingers for grasping, while negative pressure (vacuum) is needed to remove the air and return the actuator to its original position. A sensitive in-line pressure gauge can further improve the system to accurately map the internal pressure to the corresponding degree of flexion.

4.2 Modular Fingers

Another important challenge during the hardware development was the design of the air inlet connection. Initially, a brass adapter was embedded directly into the silicone base of the actuator during moulding, which has a hose-barb design on one end and a standard hollow male screw thread on the other side, which could be used to mount onto the arm. Although this provided an airtight connection, the difference in stiffness between brass and silicone caused gradual wear at the interface during repeated actuation, leading to the loss of airtightness.

To overcome this limitation, a 3D-printed PLA female adapter was designed and co-moulded with the silicone base. This modular design provided a more secure and durable pneumatic connection while simplifying assembly and replacement of components. The improved airway connection enhanced the overall robustness of the gripper and supported future integration with robotic systems.

5 Robotic Integration and Simulation

Following the successful fabrication and testing of the soft pneumatic gripper, the next phase of the project focused on its integration with a robotic manipulation system. The objective of this phase was to combine the soft gripper's grasping capabilities and the arm's precision to perform simple pick and place tasks.

5.1 Integration with RLDP-5 Robotic Arm

The fabricated soft pneumatic gripper was designed to function as the end-effector of the RLDP-5 robotic arm. A suitable mounting interface was developed to securely attach the gripper while ensuring modularity and ease of detaching and swapping the fingers out. The robotic arm provided precise positioning and controlled movement, whereas the soft gripper adapted its shape to grasp objects of different sizes and geometries without causing damage.

5.2 LiDAR-Based Object Detection

To improve the perception capabilities of the robotic system, a Unitree L2 LiDAR sensor was incorporated into the experimental setup. High-resolution 3-D point cloud data was collected from the LiDAR, and visualised in RViz. The acquired spatial information can be utilized in the future for object detection, estimation of object position, and determination of suitable grasping locations, thereby supporting future autonomous manipulation tasks.

5.3 Simulation and System Validation

Before physical implementation, the complete robotic system, including the RLDP-5 robotic arm and L2 LiDAR, was evaluated in Gazebo. The simulation was performed to verify the kinematic behaviour of the robotic arm and assess the motion planning. The MoveIt2 ROS2 package was used to load the official RLDP5 URDF files into the simulation, and subsequently generate trajectories to specified target poses. The Unilidar L2 SDK was used to interface with the lidar and obtain point cloud data. Although the

individual components of the system were validated in simulation, time constraints prevented the complete pick-and-place pipeline from being implemented and experimentally verified on the physical hardware.

6 Conclusion

This internship provided valuable practical exposure to the field of soft robotics, particularly in the design, fabrication, and integration of a soft pneumatic gripper. The project involved studying different silicone elastomers, optimizing fabrication techniques, and understanding the influence of material properties on the performance of soft actuators. Through multiple fabrication trials, the manufacturing process was refined to minimize defects such as air bubbles and improve the overall quality of the fabricated gripper.

Initially, Siloczest Liquid Silicone Rubber (LSR) was utilized for fabrication trials and process optimization due to its cost-effectiveness. Ecoflex was selected for fabricating the final soft gripper fingers because of its superior flexibility, elasticity, and bending performance under pneumatic actuation. The implementation of pre-desiccation, controlled mixing and centrifugation techniques significantly improved the quality and consistency of the fabricated actuators.

The project also focused on improving the pneumatic actuation mechanism and hardware design by developing a more reliable modular airway connection, ensuring modularity and ease of integration. The continuous refinement of material formulation, the controlled manufacturing pipeline, and the designing of pneumatic actuation contributed to the development of a reliable and efficient soft robotic gripper suitable for adaptive object manipulation.

Furthermore, the integration of the fabricated soft pneumatic gripper with the RLDP-5 robotic arm and the incorporation of the Unitree L2 LiDAR sensor demonstrated the potential of combining compliant soft robotic systems with precise robotic manipulation and intelligent sensing. This integrated approach provides a strong foundation for developing adaptive robotic grasping systems capable of safely handling delicate and irregularly shaped objects in research and industrial applications. However, due to time restrictions, we were only able to implement the individual components of the system in simulation and were unable to experimentally verify it on physical hardware.

Overall, this internship enhanced our understanding of soft materials, fabrication processes, pneumatic actuation, robotic integration, and experimental research methodologies. The knowledge and practical experience gained through this project provide a strong foundation for future work in soft robotics, biomedical engineering, industrial automation, and intelligent robotic systems.

7 Future Scope and Applications

7.1 Future Scope

The present work demonstrates the successful fabrication and integration of a soft pneumatic gripper with a robotic arm, however there is scope for further improvement. Future work could focus on obtaining pressure-flexion characteristics of the actuator, implementing closed-loop pressure control for more precise actuation, and fatigue-testing the actuator. Transferring the results from simulation to physical hardware still remains.

Further, deep learning-based algorithms could be used for object detection and advanced collision-avoidance trajectory planning could be implemented.

7.2 Applications

Soft pneumatic grippers have significant potential in applications that require safe and gentle handling of delicate objects. They can be used in biomedical engineering, healthcare, food handling, industrial automation, and agriculture. Their flexibility and adaptability make them suitable for grasping fragile and irregularly shaped objects without causing damage, making them a promising solution for future robotic systems.

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Note: An archive of the project, including CAD models, images, videos, and other supporting documents, has been uploaded to the following Google Drive folder:

https://drive.google.com/drive/folders/1_vV49bq9rFtDkehzW2WpUkA-ADjUuXYB?usp=sharing

Code files can be found at the following GitHub repository:

https://github.com/FORTFANOP/rldp5-soft-gripper